A Boundary-aware Quaternion Framework for Color Image Deblurring under Unknown Boundaries and Incomplete Observations (Supplementary Material)

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Appendix A. Proof of Theorem 1

Theorem 1 Let $\left\{\dot{\mathbf{X}}^{(k)}, \dot{\mathbf{Z}}^{(k)}, \dot{\mathbf{E}}^{(k)}, \dot{\mathbf{\Lambda}}_1^{(k)}, \dot{\mathbf{\Lambda}}_2^{(k)}\right\}$ be the sequence generated by BAQF-QWNNM. Assume that the Lagrange multiplier sequences are bounded. That is, there exist constants $M_1, M_2 > 0$ such that $\|\dot{\mathbf{\Lambda}}_1^{(k)}\|_F \leq M_1, \|\dot{\mathbf{\Lambda}}_2^{(k)}\|_F \leq M_2$, for all $k \geq 0$. Then the primal residual sequences are absolutely summable:

$$\sum_{k=0}^{\infty}\left(\left\|\dot{\mathbf{X}}^{(k+1)}-\dot{\mathbf{Z}}^{(k+1)}\right\|_F+\left\|\mathcal{H}\left(\dot{\mathbf{X}}^{(k+1)}\right)-\dot{\mathbf{E}}^{(k+1)}\right\|_F\right)<+\infty. \tag{A.1}$$

Moreover, all primal sequences have finite length:

$$\sum_{k=0}^{\infty}\left(\left\|\dot{\mathbf{X}}^{(k+1)}-\dot{\mathbf{X}}^{(k)}\right\|_F+\left\|\dot{\mathbf{Z}}^{(k+1)}-\dot{\mathbf{Z}}^{(k)}\right\|_F+\left\|\dot{\mathbf{E}}^{(k+1)}-\dot{\mathbf{E}}^{(k)}\right\|_F\right)<+\infty. \tag{A.2}$$

Consequently, there exists $\dot{\mathbf{X}}^{\star} \in \mathbb{H}^{m\times n}$ such that

$$\dot{\mathbf{X}}^{(k)} \longrightarrow \dot{\mathbf{X}}^{\star}, \quad \dot{\mathbf{Z}}^{(k)} \longrightarrow \dot{\mathbf{X}}^{\star}, \quad \dot{\mathbf{E}}^{(k)} \longrightarrow \mathcal{H}\left(\dot{\mathbf{X}}^{\star}\right). \tag{A.3}$$

Proof. For $k \geq 0$, define the two primal residuals by

$$\dot{\mathbf{R}}_1^{(k)} = \dot{\mathbf{X}}^{(k)} - \dot{\mathbf{Z}}^{(k)}, \tag{A.4}$$

and

$$\dot{\mathbf{R}}_2^{(k)} = \mathcal{H}\left(\dot{\mathbf{X}}^{(k)}\right) - \dot{\mathbf{E}}^{(k)}. \tag{A.5}$$

According to the multiplier updates,

$$\dot{\mathbf{\Lambda}}_1^{(k+1)} - \dot{\mathbf{\Lambda}}_1^{(k)} = \mu_1^{(k)} \dot{\mathbf{R}}_1^{(k+1)}, \tag{A.6}$$

and

$$\dot{\mathbf{\Lambda}}_2^{(k+1)} - \dot{\mathbf{\Lambda}}_2^{(k)} = \mu_2^{(k)} \dot{\mathbf{R}}_2^{(k+1)}. \tag{A.7}$$

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Equivalently,

$$\dot{\mathbf{R}}_1^{(k+1)} = \frac{\dot{\mathbf{\Lambda}}_1^{(k+1)} - \dot{\mathbf{\Lambda}}_1^{(k)}}{\mu_1^{(k)}}, \tag{A.8}$$

and

$$\dot{\mathbf{R}}_2^{(k+1)} = \frac{\dot{\mathbf{\Lambda}}_2^{(k+1)} - \dot{\mathbf{\Lambda}}_2^{(k)}}{\mu_2^{(k)}}. \tag{A.9}$$

Since $\|\dot{\mathbf{\Lambda}}_1^{(k)}\|_F \leq M_1, \|\dot{\mathbf{\Lambda}}_2^{(k)}\|_F \leq M_2$, for all $k \geq 0$. It obtains

$$\left\|\dot{\mathbf{R}}_1^{(k+1)}\right\|_F \leq \frac{\left\|\dot{\mathbf{\Lambda}}_1^{(k+1)}\right\|_F + \left\|\dot{\mathbf{\Lambda}}_1^{(k)}\right\|_F}{\mu_1^{(k)}} \leq \frac{2M_1}{\mu_1^{(k)}}. \tag{A.10}$$

and

$$\left\|\dot{\mathbf{R}}_2^{(k+1)}\right\|_F \leq \frac{\left\|\dot{\mathbf{\Lambda}}_2^{(k+1)}\right\|_F + \left\|\dot{\mathbf{\Lambda}}_2^{(k)}\right\|_F}{\mu_2^{(k)}} \leq \frac{2M_2}{\mu_2^{(k)}}. \tag{A.11}$$

The continuation rule gives

$$\mu_i^{(k)} = \rho^k \mu_i^{(0)}, \qquad i = 1, 2. \tag{A.12}$$

Since $\rho > 1$ and $\mu_i^{(0)} > 0$, we have

$$\sum_{k=0}^{\infty} \frac{1}{\mu_i^{(k)}} = \frac{1}{\mu_i^{(0)}} \sum_{k=0}^{\infty} \rho^{-k} = \frac{\rho}{\mu_i^{(0)}(\rho - 1)} < +\infty. \tag{A.13}$$

Therefore,

$$\sum_{k=0}^{\infty} \left\|\dot{\mathbf{R}}_1^{(k+1)}\right\|_F < +\infty, \tag{A.14}$$

and

$$\sum_{k=0}^{\infty} \left\|\dot{\mathbf{R}}_2^{(k+1)}\right\|_F < +\infty. \tag{A.15}$$

In particular,

$$\dot{\mathbf{R}}_1^{(k)} \longrightarrow \dot{\mathbf{0}}, \qquad \dot{\mathbf{R}}_2^{(k)} \longrightarrow \dot{\mathbf{0}}. \tag{A.16}$$

We next estimate the successive differences of $\{\dot{\mathbf{X}}^{(k)}\}$. According to the first-order optimality condition of $\dot{\mathbf{X}}$-subproblem, we have

$$\left(\mu_2^{(k)}\mathcal{H}^*\mathcal{H} + \mu_1^{(k)}\mathcal{I}\right)\dot{\mathbf{X}}^{(k+1)} = \mu_1^{(k)}\dot{\mathbf{Z}}^{(k)} - \dot{\mathbf{\Lambda}}_1^{(k)} + \mu_2^{(k)}\mathcal{H}^*\left(\dot{\mathbf{E}}^{(k)}\right) - \mathcal{H}^*\left(\dot{\mathbf{\Lambda}}_2^{(k)}\right). \tag{A.17}$$

By the definitions of the residuals,

$$\dot{\mathbf{Z}}^{(k)} = \dot{\mathbf{X}}^{(k)} - \dot{\mathbf{R}}_1^{(k)}, \tag{A.18}$$

and

$$\dot{\mathbf{E}}^{(k)} = \mathcal{H}\left(\dot{\mathbf{X}}^{(k)}\right) - \dot{\mathbf{R}}_2^{(k)}. \tag{A.19}$$

Substituting these identities into (A.17) yields

$$\left(\mu_2^{(k)}\mathcal{H}^*\mathcal{H} + \mu_1^{(k)}\mathcal{I}\right)\left(\dot{\mathbf{X}}^{(k+1)} - \dot{\mathbf{X}}^{(k)}\right) = -\mu_1^{(k)}\dot{\mathbf{R}}_1^{(k)} - \dot{\mathbf{\Lambda}}_1^{(k)} - \mu_2^{(k)}\mathcal{H}^*\left(\dot{\mathbf{R}}_2^{(k)}\right) - \mathcal{H}^*\left(\dot{\mathbf{\Lambda}}_2^{(k)}\right). \tag{A.20}$$

For $k \geq 1$, the multiplier updates and the continuation rule imply

$$\begin{aligned} \mu_1^{(k)} \dot{\mathbf{R}}_1^{(k)} &= \frac{\mu_1^{(k)}}{\mu_1^{(k-1)}} \left(\dot{\mathbf{\Lambda}}_1^{(k)} - \dot{\mathbf{\Lambda}}_1^{(k-1)} \right) \\ &= \rho \left(\dot{\mathbf{\Lambda}}_1^{(k)} - \dot{\mathbf{\Lambda}}_1^{(k-1)} \right), \end{aligned} \tag{A.21}$$

and

$$\begin{aligned} \mu_2^{(k)} \dot{\mathbf{R}}_2^{(k)} &= \frac{\mu_2^{(k)}}{\mu_2^{(k-1)}} \left(\dot{\mathbf{\Lambda}}_2^{(k)} - \dot{\mathbf{\Lambda}}_2^{(k-1)} \right) \\ &= \rho \left(\dot{\mathbf{\Lambda}}_2^{(k)} - \dot{\mathbf{\Lambda}}_2^{(k-1)} \right). \end{aligned} \tag{A.22}$$

Consequently,

$$\left\| \mu_1^{(k)} \dot{\mathbf{R}}_1^{(k)} \right\|_F \leq 2\rho M_1, \tag{A.23}$$

and

$$\left\| \mu_2^{(k)} \dot{\mathbf{R}}_2^{(k)} \right\|_F \leq 2\rho M_2. \tag{A.24}$$

Let $\|\mathcal{H}\|$ denote the operator norm induced by the Frobenius norm. Since $\mathcal{H}$ is bounded and $\|\mathcal{H}^*\| = \|\mathcal{H}\|$, the right-hand side of (A.20) satisfies

$$\left\| -\mu_1^{(k)} \dot{\mathbf{R}}_1^{(k)} - \dot{\mathbf{\Lambda}}_1^{(k)} - \mu_2^{(k)} \mathcal{H}^* \left(\dot{\mathbf{R}}_2^{(k)} \right) - \mathcal{H}^* \left(\dot{\mathbf{\Lambda}}_2^{(k)} \right) \right\|_F \leq 2\rho M_1 + M_1 + 2\rho \|\mathcal{H}\| M_2 + \|\mathcal{H}\| M_2 = (2\rho+1) M_1 + (2\rho+1) \|\mathcal{H}\| M_2. \tag{A.25}$$

Let $C_X = (2\rho + 1)M_1 + (2\rho + 1)\|\mathcal{H}\|M_2$.

For any compatible quaternion matrix $\dot{\mathbf{Q}}$, we have

$$\left\langle \dot{\mathbf{Q}}, \left(\mu_2^{(k)} \mathcal{H}^* \mathcal{H} + \mu_1^{(k)} \mathcal{I} \right) \dot{\mathbf{Q}} \right\rangle = \mu_2^{(k)} \left\| \mathcal{H} \left(\dot{\mathbf{Q}} \right) \right\|_F^2 + \mu_1^{(k)} \left\| \dot{\mathbf{Q}} \right\|_F^2 \geq \mu_1^{(k)} \left\| \dot{\mathbf{Q}} \right\|_F^2. \tag{A.26}$$

Therefore, the operator is positive definite and

$$\left\| \left(\mu_2^{(k)} \mathcal{H}^* \mathcal{H} + \mu_1^{(k)} \mathcal{I} \right)^{-1} \right\| \leq \frac{1}{\mu_1^{(k)}}. \tag{A.27}$$

Applying this estimate to (A.20) gives

$$\left\| \dot{\mathbf{X}}^{(k+1)} - \dot{\mathbf{X}}^{(k)} \right\|_F \leq \frac{C_X}{\mu_1^{(k)}}, \qquad k \geq 1. \tag{A.28}$$

Therefore,

$$\sum_{k=1}^{\infty} \left\| \dot{\mathbf{X}}^{(k+1)} - \dot{\mathbf{X}}^{(k)} \right\|_F \leq C_X \sum_{k=1}^{\infty} \frac{1}{\mu_1^{(k)}} < +\infty. \tag{A.29}$$

Adding finitely many initial terms does not affect summability, and hence

$$\sum_{k=0}^{\infty} \left\| \dot{\mathbf{X}}^{(k+1)} - \dot{\mathbf{X}}^{(k)} \right\|_F < +\infty. \tag{A.30}$$

Thus, $\{\dot{\mathbf{X}}^{(k)}\}$ is a Cauchy sequence. Since the quaternion matrix space is finite-dimensional and complete, there exists $\dot{\mathbf{X}}^\star \in \mathbb{H}^{m \times n}$ such that

$$\dot{\mathbf{X}}^{(k)} \longrightarrow \dot{\mathbf{X}}^\star. \tag{A.31}$$

Furthermore, since

$$\dot{\mathbf{Z}}^{(k)} = \dot{\mathbf{X}}^{(k)} - \dot{\mathbf{R}}_1^{(k)}, \tag{A.32}$$

the triangle inequality gives

$$\left\|\dot{\mathbf{Z}}^{(k+1)}-\dot{\mathbf{Z}}^{(k)}\right\|_F \leq \left\|\dot{\mathbf{X}}^{(k+1)}-\dot{\mathbf{X}}^{(k)}\right\|_F + \left\|\dot{\mathbf{R}}_1^{(k+1)}\right\|_F + \left\|\dot{\mathbf{R}}_1^{(k)}\right\|_F. \tag{A.33}$$

The three terms on the right-hand side are absolutely summable. Consequently,

$$\sum_{k=0}^{\infty}\left\|\dot{\mathbf{Z}}^{(k+1)}-\dot{\mathbf{Z}}^{(k)}\right\|_F < +\infty. \tag{A.34}$$

Similarly, since

$$\dot{\mathbf{E}}^{(k)} = \mathcal{H}\left(\dot{\mathbf{X}}^{(k)}\right) - \dot{\mathbf{R}}_2^{(k)}, \tag{A.35}$$

we have

$$\left\|\dot{\mathbf{E}}^{(k+1)}-\dot{\mathbf{E}}^{(k)}\right\|_F \leq \|\mathcal{H}\| \left\|\dot{\mathbf{X}}^{(k+1)}-\dot{\mathbf{X}}^{(k)}\right\|_F + \left\|\dot{\mathbf{R}}_2^{(k+1)}\right\|_F + \left\|\dot{\mathbf{R}}_2^{(k)}\right\|_F. \tag{A.36}$$

Again, the terms on the right-hand side are absolutely summable, and therefore

$$\sum_{k=0}^{\infty}\left\|\dot{\mathbf{E}}^{(k+1)}-\dot{\mathbf{E}}^{(k)}\right\|_F < +\infty. \tag{A.37}$$

Finally, since

$$\dot{\mathbf{R}}_1^{(k)} \longrightarrow \dot{\mathbf{0}}, \qquad \dot{\mathbf{R}}_2^{(k)} \longrightarrow \dot{\mathbf{0}}, \tag{A.38}$$

we obtain

$$\dot{\mathbf{Z}}^{(k)} = \dot{\mathbf{X}}^{(k)} - \dot{\mathbf{R}}_1^{(k)}, \quad \text{and} \quad \dot{\mathbf{Z}}^{(k)} \longrightarrow \dot{\mathbf{X}}^{\star}. \tag{A.39}$$

Moreover, the bounded linearity of $\mathcal{H}$ implies its continuity. Hence,

$$\mathcal{H}\left(\dot{\mathbf{X}}^{(k)}\right) \longrightarrow \mathcal{H}\left(\dot{\mathbf{X}}^{\star}\right). \tag{A.40}$$

It follows that

$$\dot{\mathbf{E}}^{(k)} = \mathcal{H}\left(\dot{\mathbf{X}}^{(k)}\right) - \dot{\mathbf{R}}_2^{(k)}, \quad \text{and} \quad \dot{\mathbf{E}}^{(k)} \longrightarrow \mathcal{H}\left(\dot{\mathbf{X}}^{\star}\right). \tag{A.41}$$

This completes the proof. □

Appendix B. Proof of Theorem 2

Theorem 2 Let $\left\{\dot{\mathbf{X}}^{(k)}, \dot{\mathbf{Z}}^{(k)}, \dot{\mathbf{E}}^{(k)}, \mathcal{M}^{(k)}, \dot{\mathbf{\Lambda}}_1^{(k)}, \dot{\mathbf{\Lambda}}_2^{(k)}, \mathcal{N}^{(k)}\right\}$ be the sequence generated by PnP-BAQF-QWNNM. Suppose that Assumption 1 holds and that the first two Lagrange multiplier sequences are bounded. That is, there exist constants $M_1, M_2 > 0$ such that $\|\dot{\mathbf{\Lambda}}_1^{(k)}\|_F \leq M_1, \|\dot{\mathbf{\Lambda}}_2^{(k)}\|_F \leq M_2$, for all $k \geq 0$. Then the three primal residual sequences are absolutely summable:

$$\sum_{k=0}^{\infty}\left(\left\|\dot{\mathbf{X}}^{(k+1)}-\dot{\mathbf{Z}}^{(k+1)}\right\|_F + \left\|\mathcal{H}\left(\dot{\mathbf{X}}^{(k+1)}\right)-\dot{\mathbf{E}}^{(k+1)}\right\|_F + \left\|\Gamma^{-1}\left(\dot{\mathbf{X}}^{(k+1)}\right)-\mathcal{M}^{(k+1)}\right\|_F\right) < +\infty. \tag{B.1}$$

Moreover, all primal sequences have finite length:

$$\sum_{k=0}^{\infty}\left(\left\|\dot{\mathbf{X}}^{(k+1)}-\dot{\mathbf{X}}^{(k)}\right\|_F + \left\|\dot{\mathbf{Z}}^{(k+1)}-\dot{\mathbf{Z}}^{(k)}\right\|_F + \left\|\dot{\mathbf{E}}^{(k+1)}-\dot{\mathbf{E}}^{(k)}\right\|_F + \left\|\mathcal{M}^{(k+1)}-\mathcal{M}^{(k)}\right\|_F\right) < +\infty. \tag{B.2}$$

In addition, the denoising perturbation and the scaled third multiplier are absolutely summable:

$$\sum_{k=0}^{\infty}\left\|\mathcal{M}^{(k+1)} - \left[\Gamma^{-1}\left(\dot{\mathbf{X}}^{(k+1)}\right) + \frac{\mathcal{N}^{(k)}}{\mu_3^{(k)}}\right]\right\|_F < +\infty, \quad \text{and} \quad \sum_{k=0}^{\infty}\left\|\frac{\mathcal{N}^{(k+1)}}{\mu_3^{(k)}}\right\|_F < +\infty. \tag{B.3}$$

Consequently, there exists $\dot{\mathbf{X}}^{\star} \in \mathbb{H}^{m\times n}$ such that

$$\dot{\mathbf{X}}^{(k)} \longrightarrow \dot{\mathbf{X}}^{\star}, \qquad \dot{\mathbf{Z}}^{(k)} \longrightarrow \dot{\mathbf{X}}^{\star}, \qquad \dot{\mathbf{E}}^{(k)} \longrightarrow \mathcal{H}\left(\dot{\mathbf{X}}^{\star}\right), \qquad \mathcal{M}^{(k)} \longrightarrow \Gamma^{-1}\left(\dot{\mathbf{X}}^{\star}\right). \tag{B.4}$$

Proof. Define the input to the denoiser at the $(k+1)$-th iteration by

$$\mathcal{V}^{(k)} = \Gamma^{-1}\left(\dot{\mathbf{X}}^{(k+1)}\right) + \frac{\mathcal{N}^{(k)}}{\mu_3^{(k)}}, \tag{B.5}$$

and define the corresponding denoising perturbation by

$$\mathcal{Q}^{(k+1)} = \mathcal{M}^{(k+1)} - \mathcal{V}^{(k)}. \tag{B.6}$$

According to the PnP update,

$$\mathcal{M}^{(k+1)} = \mathfrak{D}\left(\mathcal{V}^{(k)}, \delta_k\right), \qquad \delta_k = \sqrt{\frac{\gamma}{\mu_3^{(k)}}}. \tag{B.7}$$

It follows from Assumption 1 that

$$\frac{1}{3mn}\left\|\mathcal{Q}^{(k+1)}\right\|_F^2 \leq C\delta_k^2 = \frac{C\gamma}{\mu_3^{(k)}}. \tag{B.8}$$

Let

$$C_{\mathfrak{D}} = \sqrt{3mnC\gamma}. \tag{B.9}$$

Then

$$\left\|\mathcal{Q}^{(k+1)}\right\|_F \leq \frac{C_{\mathfrak{D}}}{\sqrt{\mu_3^{(k)}}}. \tag{B.10}$$

Since

$$\mu_3^{(k)} = \rho^k \mu_3^{(0)}, \tag{B.11}$$

we have

$$\sum_{k=0}^{\infty} \frac{1}{\sqrt{\mu_3^{(k)}}} = \frac{1}{\sqrt{\mu_3^{(0)}}} \sum_{k=0}^{\infty} \rho^{-k/2} < +\infty. \tag{B.12}$$

Therefore,

$$\sum_{k=0}^{\infty} \left\|\mathcal{Q}^{(k+1)}\right\|_F < +\infty. \tag{B.13}$$

By the update of the third multiplier,

$$\begin{aligned}
\mathcal{N}^{(k+1)} &= \mathcal{N}^{(k)} + \mu_3^{(k)}\left[\Gamma^{-1}\left(\dot{\mathbf{X}}^{(k+1)}\right) - \mathcal{M}^{(k+1)}\right] \\
&= \mu_3^{(k)}\left[\mathcal{V}^{(k)} - \mathcal{M}^{(k+1)}\right] \\
&= -\mu_3^{(k)}\mathcal{Q}^{(k+1)}.
\end{aligned} \tag{B.14}$$

Consequently,

$$\left\|\mathcal{N}^{(k+1)}\right\|_F \leq C_{\mathfrak{D}}\sqrt{\mu_3^{(k)}} \tag{B.15}$$

and

$$\frac{\mathcal{N}^{(k+1)}}{\mu_3^{(k)}} = -\mathcal{Q}^{(k+1)}. \tag{B.16}$$

Together with (B.13), this gives

$$\sum_{k=0}^{\infty} \left\|\frac{\mathcal{N}^{(k+1)}}{\mu_3^{(k)}}\right\|_F < +\infty. \tag{B.17}$$

For $k \geq 0$, define the three primal residuals by

$$\dot{\mathbf{R}}_1^{(k)} = \dot{\mathbf{X}}^{(k)} - \dot{\mathbf{Z}}^{(k)}, \tag{B.18}$$

$$\dot{\mathbf{R}}_2^{(k)} = \mathcal{H}\left(\dot{\mathbf{X}}^{(k)}\right) - \dot{\mathbf{E}}^{(k)}, \tag{B.19}$$

and

$$\mathcal{R}_3^{(k)} = \Gamma^{-1}\left(\dot{\mathbf{X}}^{(k)}\right) - \mathcal{M}^{(k)}. \tag{B.20}$$

According to the first two multiplier updates,

$$\dot{\mathbf{\Lambda}}_1^{(k+1)} - \dot{\mathbf{\Lambda}}_1^{(k)} = \mu_1^{(k)} \dot{\mathbf{R}}_1^{(k+1)} \tag{B.21}$$

and

$$\dot{\mathbf{\Lambda}}_2^{(k+1)} - \dot{\mathbf{\Lambda}}_2^{(k)} = \mu_2^{(k)} \dot{\mathbf{R}}_2^{(k+1)}. \tag{B.22}$$

Therefore,

$$\left\|\dot{\mathbf{R}}_1^{(k+1)}\right\|_F \leq \frac{2M_1}{\mu_1^{(k)}}, \qquad \left\|\dot{\mathbf{R}}_2^{(k+1)}\right\|_F \leq \frac{2M_2}{\mu_2^{(k)}}. \tag{B.23}$$

Since

$$\mu_i^{(k)} = \rho^k \mu_i^{(0)}, \qquad i = 1, 2, \tag{B.24}$$

we obtain

$$\sum_{k=0}^{\infty} \left\|\dot{\mathbf{R}}_1^{(k+1)}\right\|_F < +\infty, \qquad \sum_{k=0}^{\infty} \left\|\dot{\mathbf{R}}_2^{(k+1)}\right\|_F < +\infty. \tag{B.25}$$

For the third primal residual, the definitions of $\mathcal{V}^{(k)}$ and $\mathcal{Q}^{(k+1)}$ imply

$$\begin{aligned}\mathcal{R}_3^{(k+1)} &= \Gamma^{-1}\left(\dot{\mathbf{X}}^{(k+1)}\right) - \mathcal{M}^{(k+1)} \\ &= -\mathcal{Q}^{(k+1)} - \frac{\mathcal{N}^{(k)}}{\mu_3^{(k)}}.\end{aligned} \tag{B.26}$$

For $k \geq 1$, (B.15) gives

$$\left\|\mathcal{N}^{(k)}\right\|_F \leq C_{\mathfrak{D}} \sqrt{\mu_3^{(k-1)}}. \tag{B.27}$$

Hence,

$$\begin{aligned}\left\|\mathcal{R}_3^{(k+1)}\right\|_F &\leq \frac{C_{\mathfrak{D}}}{\sqrt{\mu_3^{(k)}}} + \frac{C_{\mathfrak{D}}\sqrt{\mu_3^{(k-1)}}}{\mu_3^{(k)}} \\ &= \frac{C_{\mathfrak{D}}\left(1 + \rho^{-1/2}\right)}{\sqrt{\mu_3^{(k)}}}.\end{aligned} \tag{B.28}$$

The initial term is finite and does not affect summability. Therefore,

$$\sum_{k=0}^{\infty} \left\|\mathcal{R}_3^{(k+1)}\right\|_F < +\infty. \tag{B.29}$$

Thus, the three primal residual sequences are absolutely summable. In particular,

$$\dot{\mathbf{R}}_1^{(k)} \longrightarrow \dot{\mathbf{0}}, \qquad \dot{\mathbf{R}}_2^{(k)} \longrightarrow \dot{\mathbf{0}}, \qquad \left\|\mathcal{R}_3^{(k)}\right\|_F \longrightarrow 0. \tag{B.30}$$

We next establish the finite-length property of $\{\dot{\mathbf{X}}^{(k)}\}$. Under the RGB-quaternion correspondence defined by Γ, the mappings Γ and Γ^{-1} are linear isometries on the adopted quaternion image space. In particular,

$$\|\Gamma(\mathcal{Q})\|_F = \|\mathcal{Q}\|_F \tag{B.31}$$

and

$$\Gamma\left(\Gamma^{-1}\left(\dot{\mathbf{Q}}\right)\right) = \dot{\mathbf{Q}} \tag{B.32}$$

for every compatible $\mathcal{Q}$ and $\dot{\mathbf{Q}}$.

The first-order optimality condition of the $\dot{\mathbf{X}}$-subproblem is

$$\left[\mu_2^{(k)}\mathcal{H}^*\mathcal{H} + \left(\mu_1^{(k)} + \mu_3^{(k)}\right)\mathcal{I}\right]\dot{\mathbf{X}}^{(k+1)} = \mu_1^{(k)}\dot{\mathbf{Z}}^{(k)} - \dot{\mathbf{\Lambda}}_1^{(k)} + \mu_2^{(k)}\mathcal{H}^*\left(\dot{\mathbf{E}}^{(k)}\right) - \mathcal{H}^*\left(\dot{\mathbf{\Lambda}}_2^{(k)}\right) + \mu_3^{(k)}\Gamma\left(\mathcal{M}^{(k)}\right) - \Gamma\left(\mathcal{N}^{(k)}\right). \tag{B.33}$$

By the definitions of the residuals,

$$\dot{\mathbf{Z}}^{(k)} = \dot{\mathbf{X}}^{(k)} - \dot{\mathbf{R}}_1^{(k)}, \tag{B.34}$$

$$\dot{\mathbf{E}}^{(k)} = \mathcal{H}\left(\dot{\mathbf{X}}^{(k)}\right) - \dot{\mathbf{R}}_2^{(k)}, \tag{B.35}$$

and

$$\mathcal{M}^{(k)} = \Gamma^{-1}\left(\dot{\mathbf{X}}^{(k)}\right) - \mathcal{R}_3^{(k)}. \tag{B.36}$$

Substituting these identities into (B.33) gives

$$\left[\mu_2^{(k)}\mathcal{H}^*\mathcal{H} + \left(\mu_1^{(k)} + \mu_3^{(k)}\right)\mathcal{I}\right]\left(\dot{\mathbf{X}}^{(k+1)} - \dot{\mathbf{X}}^{(k)}\right) = -\mu_1^{(k)}\dot{\mathbf{R}}_1^{(k)} - \dot{\mathbf{\Lambda}}_1^{(k)} - \mu_2^{(k)}\mathcal{H}^*\left(\dot{\mathbf{R}}_2^{(k)}\right) - \mathcal{H}^*\left(\dot{\mathbf{\Lambda}}_2^{(k)}\right) - \mu_3^{(k)}\Gamma\left(\mathcal{R}_3^{(k)}\right) - \Gamma\left(\mathcal{N}^{(k)}\right). \tag{B.37}$$

For $k \geq 1$, the first two multiplier updates and the continuation rule imply

$$\mu_1^{(k)}\dot{\mathbf{R}}_1^{(k)} = \rho\left(\dot{\mathbf{\Lambda}}_1^{(k)} - \dot{\mathbf{\Lambda}}_1^{(k-1)}\right) \tag{B.38}$$

and

$$\mu_2^{(k)}\dot{\mathbf{R}}_2^{(k)} = \rho\left(\dot{\mathbf{\Lambda}}_2^{(k)} - \dot{\mathbf{\Lambda}}_2^{(k-1)}\right). \tag{B.39}$$

Consequently,

$$\left\|\mu_1^{(k)}\dot{\mathbf{R}}_1^{(k)}\right\|_F \leq 2\rho M_1, \qquad \left\|\mu_2^{(k)}\dot{\mathbf{R}}_2^{(k)}\right\|_F \leq 2\rho M_2. \tag{B.40}$$

For $k \geq 2$, the third multiplier update gives

$$\mu_3^{(k)}\mathcal{R}_3^{(k)} = \rho\left(\mathcal{N}^{(k)} - \mathcal{N}^{(k-1)}\right). \tag{B.41}$$

Hence,

$$\begin{aligned}
&\left\|\mu_3^{(k)}\mathcal{R}_3^{(k)} + \mathcal{N}^{(k)}\right\|_F \\
&= \left\|(\rho+1)\,\mathcal{N}^{(k)} - \rho\mathcal{N}^{(k-1)}\right\|_F \\
&\leq (\rho+1)\left\|\mathcal{N}^{(k)}\right\|_F + \rho\left\|\mathcal{N}^{(k-1)}\right\|_F \\
&\leq C_{\mathfrak{D}}\left[(\rho+1)\sqrt{\mu_3^{(k-1)}} + \rho\sqrt{\mu_3^{(k-2)}}\right] \\
&= C_{\mathfrak{D}}\left(\frac{\rho+1}{\sqrt{\rho}} + 1\right)\sqrt{\mu_3^{(k)}}.
\end{aligned} \tag{B.42}$$

Define

$$C_0 = (2\rho+1)\,M_1 + (2\rho+1)\,\|\mathcal{H}\|M_2 \tag{B.43}$$

and

$$C_3 = C_{\mathfrak{D}} \left(\frac{\rho + 1}{\sqrt{\rho}} + 1 \right), \tag{B.44}$$

where $\|\mathcal{H}\|$ denotes the operator norm induced by the Frobenius norm. Since Γ is an isometry and $\|\mathcal{H}^*\| = \|\mathcal{H}\|$, the right-hand side (RHS) of (B.37) satisfies

$$\|\text{RHS of (B.37)}\|_F \leq C_0 + C_3 \sqrt{\mu_3^{(k)}}, \qquad k \geq 2. \tag{B.45}$$

Because $\mathcal{H}^*\mathcal{H}$ is positive semidefinite,

$$\mu_2^{(k)} \mathcal{H}^*\mathcal{H} + \left(\mu_1^{(k)} + \mu_3^{(k)} \right) \mathcal{I} \geq \left(\mu_1^{(k)} + \mu_3^{(k)} \right) \mathcal{I}. \tag{B.46}$$

Therefore,

$$\left\| \left[\mu_2^{(k)} \mathcal{H}^*\mathcal{H} + \left(\mu_1^{(k)} + \mu_3^{(k)} \right) \mathcal{I} \right]^{-1} \right\| \leq \frac{1}{\mu_1^{(k)} + \mu_3^{(k)}}. \tag{B.47}$$

It follows from (B.37) that

$$\begin{aligned} \left\| \dot{\mathbf{X}}^{(k+1)} - \dot{\mathbf{X}}^{(k)} \right\|_F &\leq \frac{C_0 + C_3 \sqrt{\mu_3^{(k)}}}{\mu_1^{(k)} + \mu_3^{(k)}} \\ &= \frac{C_0}{\mu_1^{(0)} + \mu_3^{(0)}} \rho^{-k} + \frac{C_3 \sqrt{\mu_3^{(0)}}}{\mu_1^{(0)} + \mu_3^{(0)}} \rho^{-k/2}, \qquad k \geq 2. \end{aligned} \tag{B.48}$$

Since both $\sum_{k=2}^{\infty} \rho^{-k}$ and $\sum_{k=2}^{\infty} \rho^{-k/2}$ are convergent, we obtain

$$\sum_{k=0}^{\infty} \left\| \dot{\mathbf{X}}^{(k+1)} - \dot{\mathbf{X}}^{(k)} \right\|_F < +\infty. \tag{B.49}$$

Thus, $\{\dot{\mathbf{X}}^{(k)}\}$ is a Cauchy sequence. Since the quaternion matrix space is finite-dimensional and complete, there exists $\dot{\mathbf{X}}^\star \in \mathbb{H}^{m \times n}$ such that

$$\dot{\mathbf{X}}^{(k)} \longrightarrow \dot{\mathbf{X}}^\star. \tag{B.50}$$

Furthermore,

$$\left\| \dot{\mathbf{Z}}^{(k+1)} - \dot{\mathbf{Z}}^{(k)} \right\|_F \leq \left\| \dot{\mathbf{X}}^{(k+1)} - \dot{\mathbf{X}}^{(k)} \right\|_F + \left\| \dot{\mathbf{R}}_1^{(k+1)} \right\|_F + \left\| \dot{\mathbf{R}}_1^{(k)} \right\|_F, \tag{B.51}$$

$$\left\| \dot{\mathbf{E}}^{(k+1)} - \dot{\mathbf{E}}^{(k)} \right\|_F \leq \|\mathcal{H}\| \left\| \dot{\mathbf{X}}^{(k+1)} - \dot{\mathbf{X}}^{(k)} \right\|_F + \left\| \dot{\mathbf{R}}_2^{(k+1)} \right\|_F + \left\| \dot{\mathbf{R}}_2^{(k)} \right\|_F, \tag{B.52}$$

and

$$\left\| \mathcal{M}^{(k+1)} - \mathcal{M}^{(k)} \right\|_F \leq \left\| \dot{\mathbf{X}}^{(k+1)} - \dot{\mathbf{X}}^{(k)} \right\|_F + \left\| \mathcal{R}_3^{(k+1)} \right\|_F + \left\| \mathcal{R}_3^{(k)} \right\|_F. \tag{B.53}$$

The terms on the right-hand sides are absolutely summable. Hence,

$$\sum_{k=0}^{\infty} \left(\left\| \dot{\mathbf{Z}}^{(k+1)} - \dot{\mathbf{Z}}^{(k)} \right\|_F + \left\| \dot{\mathbf{E}}^{(k+1)} - \dot{\mathbf{E}}^{(k)} \right\|_F + \left\| \mathcal{M}^{(k+1)} - \mathcal{M}^{(k)} \right\|_F \right) < +\infty. \tag{B.54}$$

Finally, since all three primal residuals converge to zero, it obtains

$$\begin{aligned} \dot{\mathbf{Z}}^{(k)} &= \dot{\mathbf{X}}^{(k)} - \dot{\mathbf{R}}_1^{(k)}, \quad \text{and} \quad \dot{\mathbf{Z}}^{(k)} \longrightarrow \dot{\mathbf{X}}^\star, \\ \dot{\mathbf{E}}^{(k)} &= \mathcal{H}\left(\dot{\mathbf{X}}^{(k)} \right) - \dot{\mathbf{R}}_2^{(k)} \quad \text{and} \quad \dot{\mathbf{E}}^{(k)} \longrightarrow \mathcal{H}\left(\dot{\mathbf{X}}^\star \right), \\ \mathcal{M}^{(k)} &= \Gamma^{-1}\left(\dot{\mathbf{X}}^{(k)} \right) - \mathcal{R}_3^{(k)} \quad \text{and} \quad \mathcal{M}^{(k)} \longrightarrow \Gamma^{-1}\left(\dot{\mathbf{X}}^\star \right). \end{aligned} \tag{B.55}$$

This completes the proof. □